EECE 306 Lab 1 notebook. Building the Toolbox, Constants and Vector Algebra

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Fill in every item marked TODO, then run `publish('Lab01_notebook.m')` from the toolbox root and print the resulting HTML to PDF. Three to five pages. The notebook is graded on required results, verification evidence, interpretation, honesty about problems, and presentation.

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Setup

This cell must run clean from the toolbox root.

```
clear; close all;
disp(version)
```

```
24.2.0.3212159 (R2024b) Update 9
```

Part I. Physical constants

The two derived constants must be computed, not typed in.

```
fprintf('eps0 = %.10e F/m\n', em.const.eps0());
fprintf('mu0  = %.10e H/m\n', em.const.mu0());
fprintf('c0   = %.10e m/s\n', em.const.c0());
fprintf('eta0 = %.6f ohm\n',  em.const.eta0());
em.test.assertClose(em.const.c0(), 1/sqrt(em.const.mu0()*em.const.eps0()), 1e-14, 'c0 derived');
em.test.assertClose(em.const.eta0(), sqrt(em.const.mu0()/em.const.eps0()), 1e-14, 'eta0 derived');
disp('Derived constant checks passed.')
```

```
eps0 = 8.8541878128e-12 F/m
mu0  = 1.2566370614e-06 H/m
c0   = 2.9979245808e+08 m/s
eta0 = 376.730314 ohm
Derived constant checks passed.
```

Part II. Vector algebra on the required checks

```
fprintf('mag([3 4 0])  = %.15g   (expect 5 exactly)\n', em.vec.mag([3 4 0]));
disp('unit([3 4 0]) ='); disp(em.vec.unit([3 4 0]))
disp('unit([0 0 0]) (expect all NaN) ='); disp(em.vec.unit([0 0 0]))
fprintf('angle perpendicular = %.15g  (expect pi/2)\n', em.vec.angle([1 0 0],[0 1 0]));
ap = em.vec.angle([1 0 0],[-1 0 0]);
fprintf('angle antiparallel  = %.15g, real = %d  (expect pi, 1)\n', ap, isreal(ap));
disp('fromTo([1 1 1],[2 3 4]) ='); disp(em.vec.fromTo([1 1 1],[2 3 4]))
```

```
mag([3 4 0])  = 5   (expect 5 exactly)
unit([3 4 0]) =
    0.6000    0.8000         0

unit([0 0 0]) (expect all NaN) =
   NaN   NaN   NaN

angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
     1     2     3
```

Part II continued. Shape discipline

Every function must accept Nx3 input and return the documented shape.

```
A = randn(100,3); B = randn(100,3);
fprintf('mag    input 100x3 output %s\n', mat2str(size(em.vec.mag(A))));
fprintf('unit   input 100x3 output %s\n', mat2str(size(em.vec.unit(A))));
fprintf('angle  input 100x3 output %s\n', mat2str(size(em.vec.angle(A,B))));
fprintf('fromTo input 100x3 output %s\n', mat2str(size(em.vec.fromTo(A,B))));
```

```
mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
```

Test Output

```
PASS  test_lab01            0.10 s
------------------------------------------------------------
1 passed, 0 failed, 1 total
All tests passing.
```

Interpretation

TODO three to six sentences in your own words. Why must c0 and eta0 be computed from eps0 and mu0 rather than typed in as numbers, and why is == the wrong way to compare a computed field value against a formula.

1. Since c0 and eta0 are constants, it is important to have as much precision as possible, especially when many significant digits are needed. Using a number instead of a function will introduce rounding errors that can significantly alter results. 2. The comparison operator "==" is not effective for comparing a computed field value against a formula because floating point numbers calculated by the system may not match the formula exactly due to error, and thus using "==" will evaluate the statement to "False."

Problems encountered

TODO state plainly anything that did not work and what was tried. An honest account of a failure earns more credit than silence. Write NONE if the session was clean.

I tried computing the interior angle by writing the equation as one line: th = arccos((A .* B) ./ (em.vec.mag(A) .* em.vec.mag(B)). This did produce a result, but the output (when testing) was a 100x3 vector instead of a 100x1 vector. I had to compute each value separately, write the cosine, clamp it, and then use arccos on that value to determine theta. I also tried running tests inside the test function, which gave me an infinite loop... it took me a bit before I realized I needed to type in the prompt in the command window. Also, this lab I learned that you need .*, ./, and .^ to multiply, divide, and raise to a power for element-by-element operations instead of *, /, and ^!

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
------------------------------------------------------------
24.2.0.3212159 (R2024b) Update 9
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Derived constant checks passed.
mag([3 4 0])  = 5   (expect 5 exactly)
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angle perpendicular = 1.5707963267949  (expect pi/2)
angle antiparallel  = 3.14159265358979, real = 1  (expect pi, 1)
fromTo([1 1 1],[2 3 4]) =
     1     2     3

mag    input 100x3 output [100 1]
unit   input 100x3 output [100 3]
angle  input 100x3 output [100 1]
fromTo input 100x3 output [100 3]
  PASS  test_lab01            0.02 s
spherical   roundtrip max error = 5.449e+00
cylindrical roundtrip max error = 5.478e+00
c2sph([0 0 1])  =
     1     0     0

c2sph([0 0 -1]) =
    1.0000    3.1416         0

phi in the four quadrants = 5.4978 all inside [0, 2*pi) = 1
vector roundtrip (spherical pair) max error = 4.940e+00
magnitude change under conversion max       = 2.837e+00
vector roundtrip (cylindrical pair) max error = 4.940e+00
at (0, 5, 0), (Ar, Atheta, Aphi) =
    3.0000    0.0000   -2.0000
```

```
at (4, 0, 0), (Ar, Atheta, Aphi) =
    2.0000    0.0000    3.0000

  PASS  test_lab02            0.03 s
   r (m)    relative error
    1.0000         0
    2.0000         0
    3.0000         0
    4.0000         0
    5.0000         0
    6.0000         0
    7.0000         0
    8.0000         0
    9.0000    0.0000
   10.0000         0

max |Ex| on bisector      = 0.000e+00 V/m
max |E| on the same plane = 6.631e+02 V/m
|E(r)| / |E(2r)| = 8.0003   (expect near 8 for a dipole)
em.field.E on 5000 points took 0.004 s  (requirement, under 2 s)
  PASS  test_lab03            0.38 s
----------------------------------------------------------------
  3 passed, 0 failed, 3 total
  All tests passing.
```

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